Computer Networks (CL3001)

CN LAB 07

Submitted by:

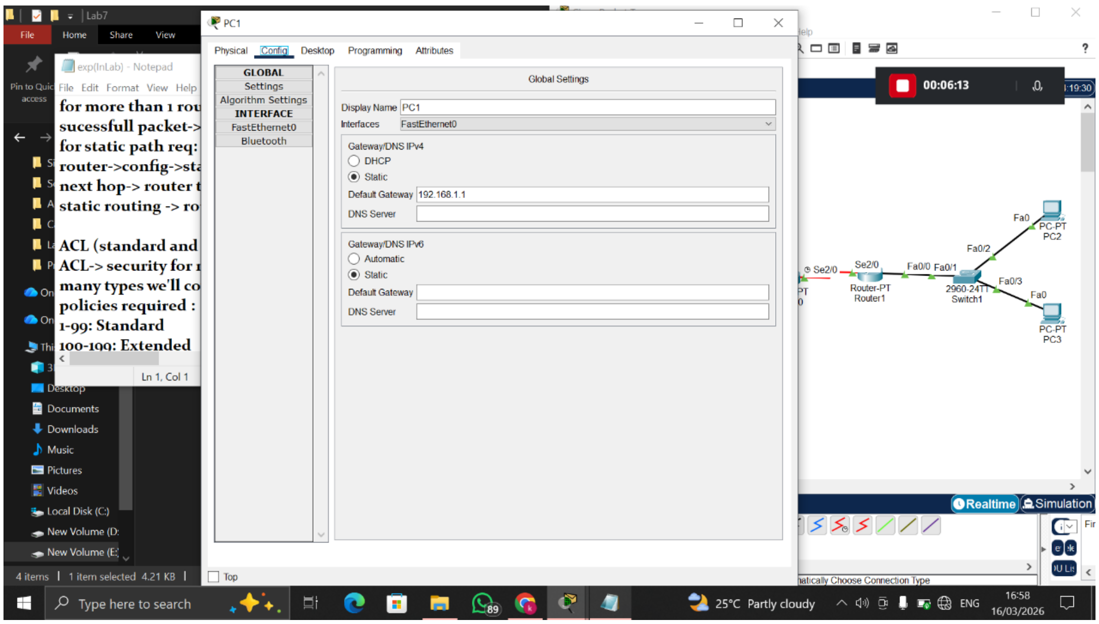
Syeda Sara Ali
23K-0070
BS-AI (6A)

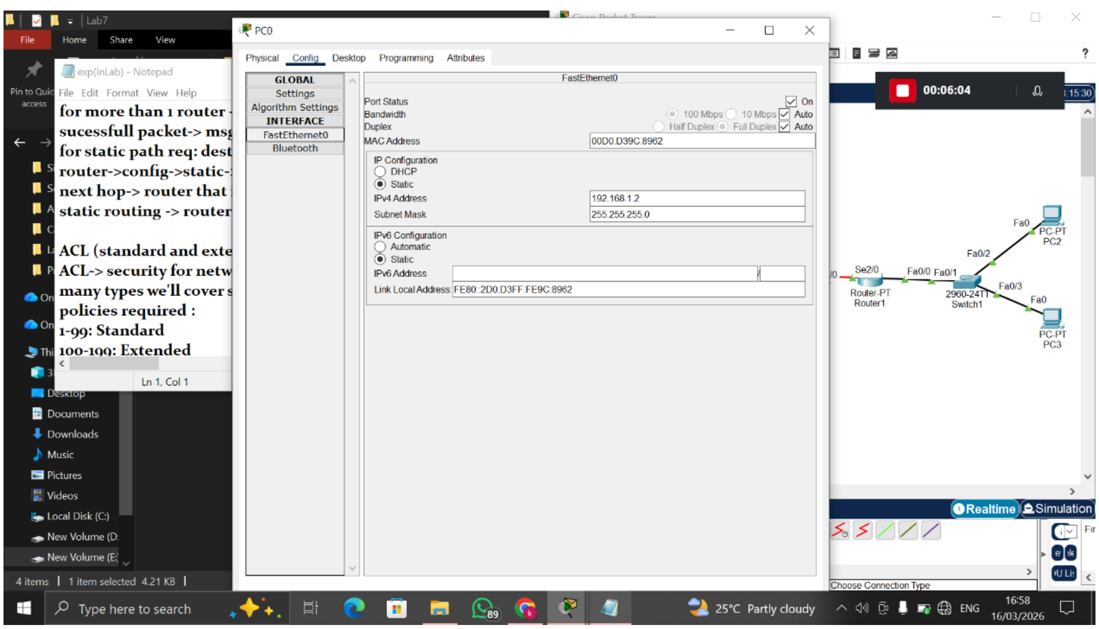
Submitted to:

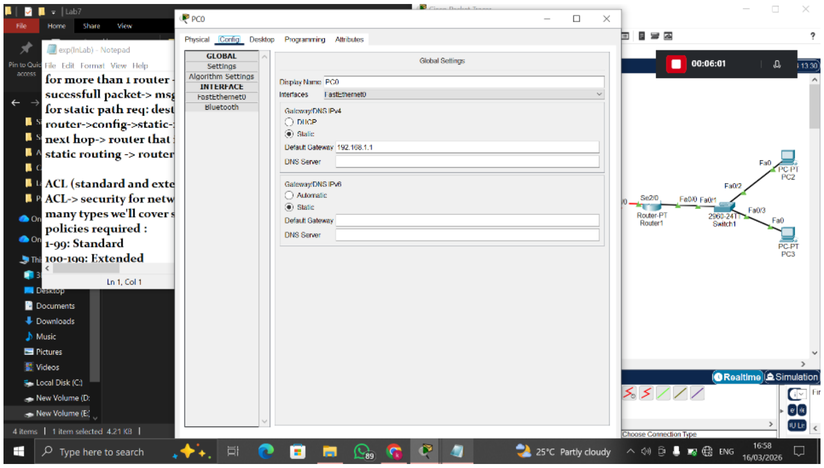
Sir Sameer Faisal

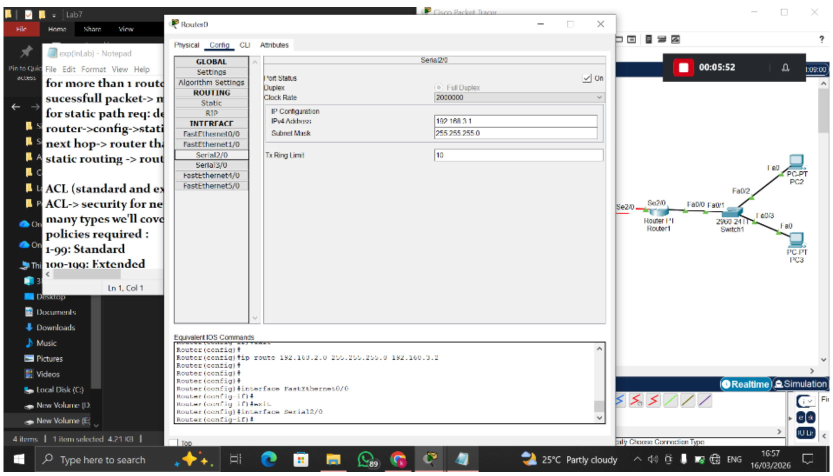
Lab Tasks:

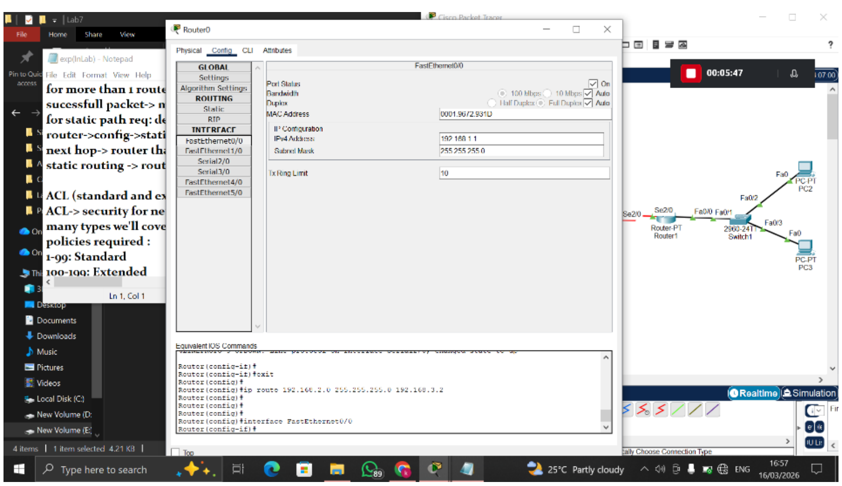
1. Prevent PCs in the STUDENTS network from communicating with any device in the TEACHERS network. But TEACHERS should be able to communicate with the STUDENTS.

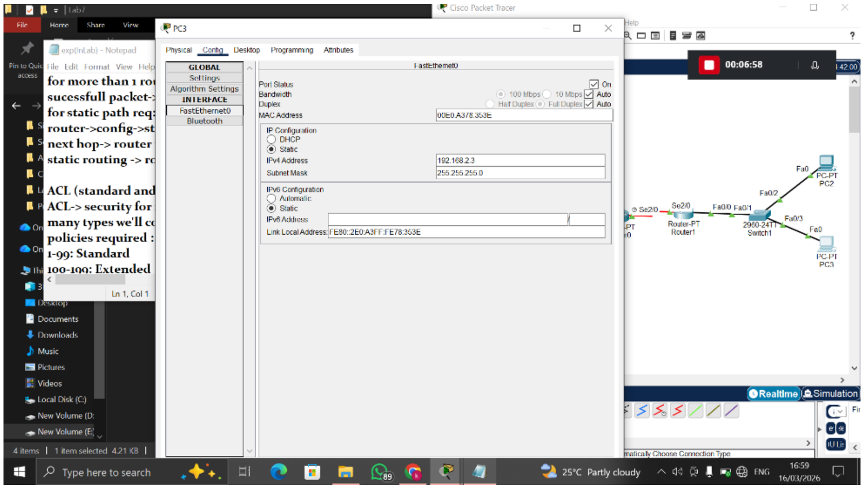


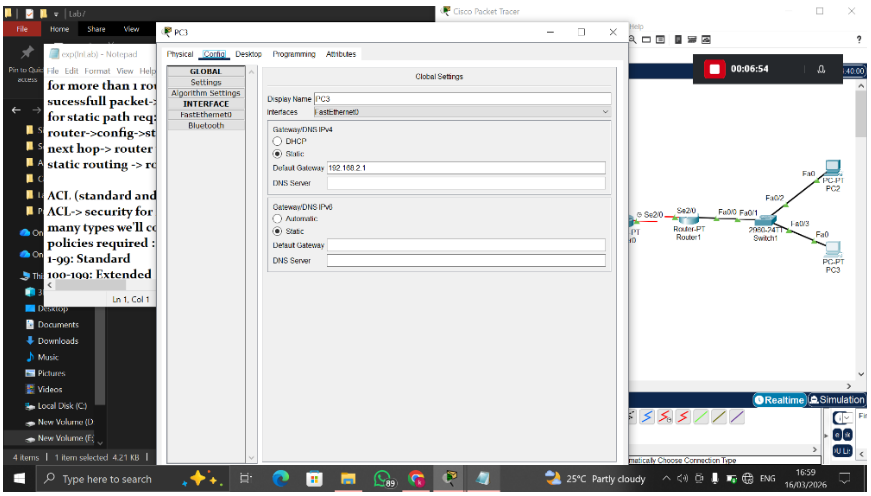


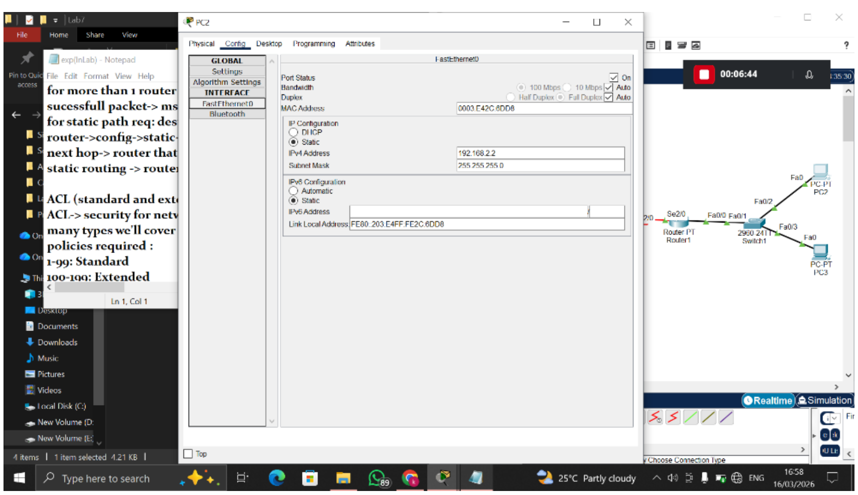

PC0
Physical Config Desktop Programming Attributes
Global Settings
Display Name PC0
Interfaces FastEthernet0
Gateway/DNS IPv4
DHCP
Static
Default Gateway 192.168.1.1
DNS Server
Gateway/DNS IPv6
Automatic
Static
Default Gateway
DNS Server
for more than 1 router
sucessfull packet-> msg
for static path req: dest
router->config->static
next hop-> router that
static routing -> router
ACL (standard and exte
ACL-> security for netw
many types we'll cover
policies required :
1-99: Standard
100-199: Extended
00:06:01
Realtime Simulation
Type here to search
25°C Partly cloudy
ENG
16:58
16/03/2026

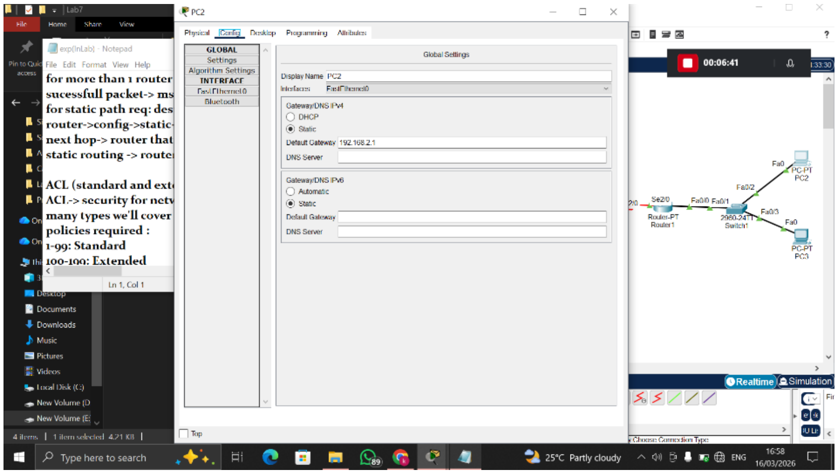

Router0
Physical Config CLI Attributes
Serial2/0
Port Status On
Duplex Full Duplex
Clock Rate 2000000
IP Configuration
IPv4 Address 192.168.3.1
Subnet Mask 255.255.255.0
Tx Ring Limit 10
Equivalent IOS Commands
00:05:52
Realtime Simulation
Type here to search
25°C Partly cloudy
ENG
16:57
16/03/2026

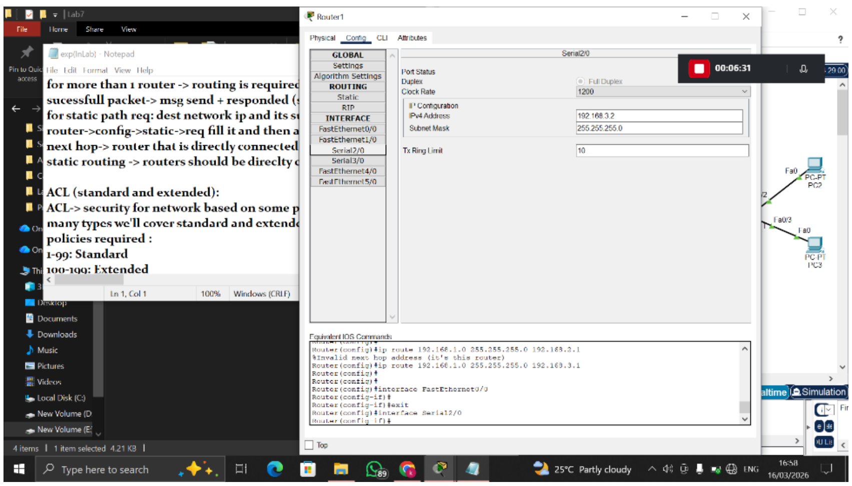

Router0
Physical Config CLI Attributes
FastEthernet0/0
Port Status On
Bandwidth 100 Mbps 10 Mbps Auto
Duplex Half Duplex Full Duplex Auto
MAC Address 0001.9672.931D
IP Configuration
IPv4 Address 192.168.1.1
Subnet Mask 255.255.255.0
Tx Ring Limit 10
Equivalent IOS Commands
00:05:47
Realtime Simulation
Type here to search
25°C Partly cloudy
ENG
16:57
16/03/2026

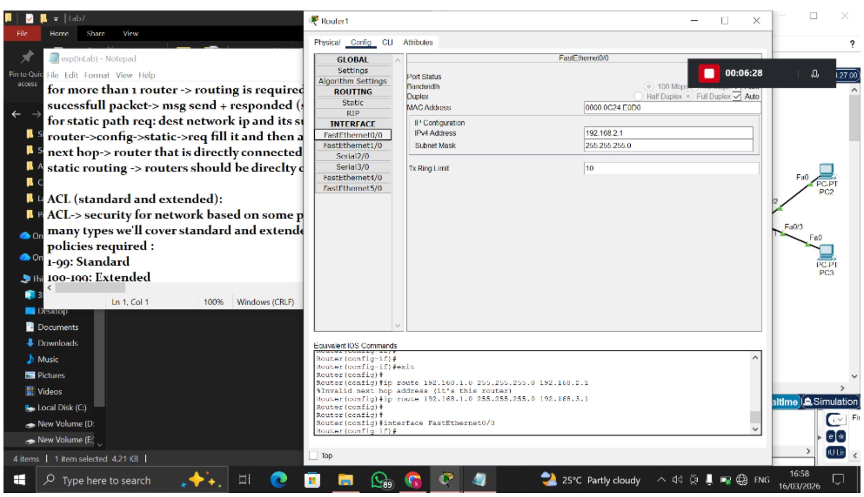

PC3
Physical
Config
Desktop
Programming
Attributes
GLOBAL
Settings
Algorithm Settings
INTERFACE
FastEthernet0
Bluetooth
FastEthernet0
Port Status
Bandwidth
Duplex
MAC Address
00E0.A378.353E
IP Configuration
DHCP
Static
IPv4 Address
192.168.2.3
Subnet Mask
255.255.255.0
IPv6 Configuration
Automatic
Static
IPv6 Address
Link Local Address
FE80::2E0:A3FF:FE78:353E
for more than 1 ro
sucessfull packet-
for static path req
router->config->st
next hop-> router
static routing -> r
ACL (standard and
ACL-> security for
many types we'll c
policies required :
1-99: Standard
100-199: Extended
Ln 1, Col 1
Cisco Packet Tracer
00:06:58
PC-PT
PC2
PC3
Router-PT
Router1
2960-24TT
Switch1
Realtime
Simulation
Type here to search
25°C Partly cloudy
ENG
16:59
16/03/2026

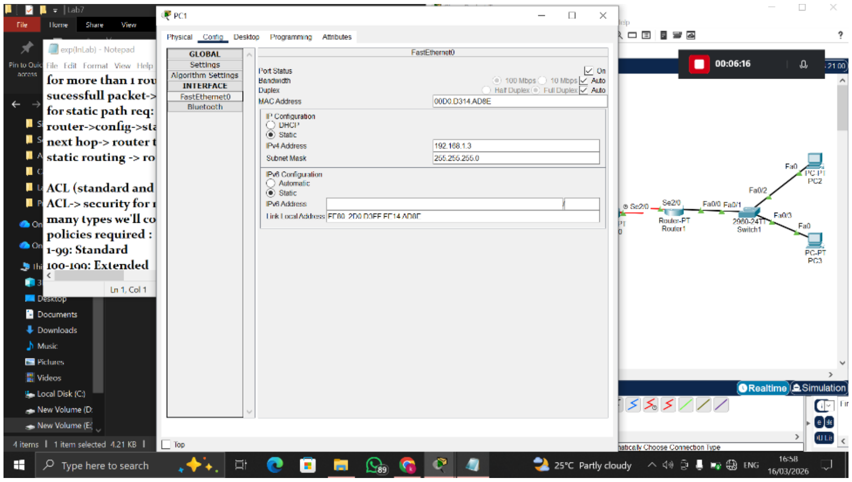

PC3
Physical
Config
Desktop
Programming
Attributes
GLOBAL
Settings
Algorithm Settings
INTERFACE
FastEthernet0
Bluetooth
Global Settings
Display Name
PC3
Interfaces
FastEthernet0
Gateway/DNS IPv4
DHCP
Static
Default Gateway
192.168.2.1
DNS Server
Gateway/DNS IPv6
Automatic
Static
Default Gateway
DNS Server
for more than 1 ro
sucessfull packet-
for static path req
router->config->st
next hop-> router
static routing -> r
ACL (standard and
ACL-> security for
many types we'll c
policies required :
1-99: Standard
100-199: Extended
Cisco Packet Tracer
00:06:54
PC-PT
PC2
PC3
Router-PT
Router1
Switch1
Realtime
Simulation
Type here to search
25°C Partly cloudy
ENG
16:59
16/03/2026

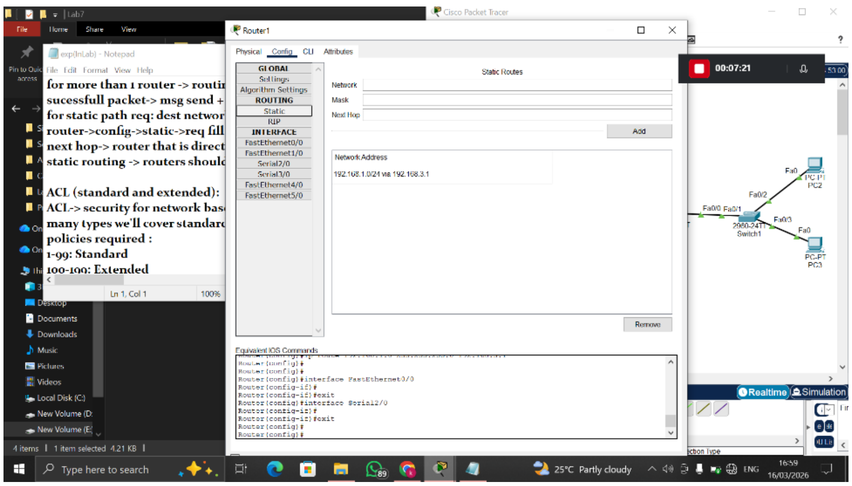

PC2
Physical
Config
Desktop
Programming
Attributes
GLOBAL
Settings
Algorithm Settings
INTERFACE
FastEthernet0
Bluetooth
FastEthernet0
Port Status
Bandwidth
Duplex
MAC Address
0003.E42C.6DD8
IP Configuration
DHCP
Static
IPv4 Address
192.168.2.2
Subnet Mask
255.255.255.0
IPv6 Configuration
Automatic
Static
IPv6 Address
Link Local Address
FE80::203:E4FF:FE2C:6DD8
Top
for more than 1 router
sucessfull packet-> ms
for static path req: des
router->config->static
next hop-> router that
static routing -> route
ACL (standard and ext
ACL-> security for netw
many types we'll cover
policies required :
1-99: Standard
100-199: Extended
00:06:44
PC-PT
PC2
PC3
Router-PT
Router1
2960-24TT
Switch1
Realtime
Simulation
Type here to search
25°C Partly cloudy
ENG
16:58
16/03/2026

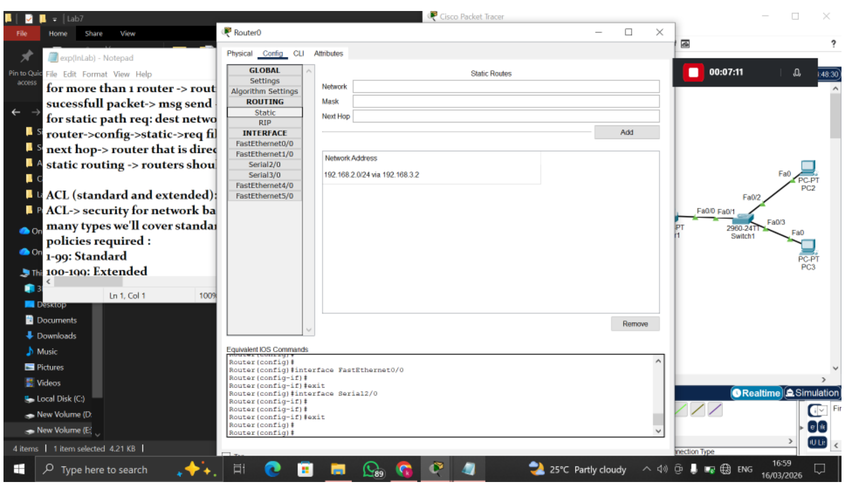

exp(InLab) - Notepad
for more than 1 router
sucessfull packet-> ms
for static path req: des
router->config->static-
next hop-> router that
static routing -> route
ACL (standard and ext
ACL-> security for netw
many types we'll cover
policies required :
1-99: Standard
100-199: Extended
PC2
Physical Config Desktop Programming Attributes
GLOBAL
Settings
Algorithm Settings
INTERFACE
FastEthernet0
Bluetooth
Global Settings
Display Name PC2
Interfaces FastEthernet0
Gateway/DNS IPv4
DHCP
Static
Default Gateway 192.168.2.1
DNS Server
Gateway/DNS IPv6
Automatic
Static
Default Gateway
DNS Server
00:06:41
Router-PT
Router1
2960-24TT
Switch1
PC-PT
PC2
PC-PT
PC3
Realtime
Simulation
Type here to search
25°C Partly cloudy
16:58
16/03/2026

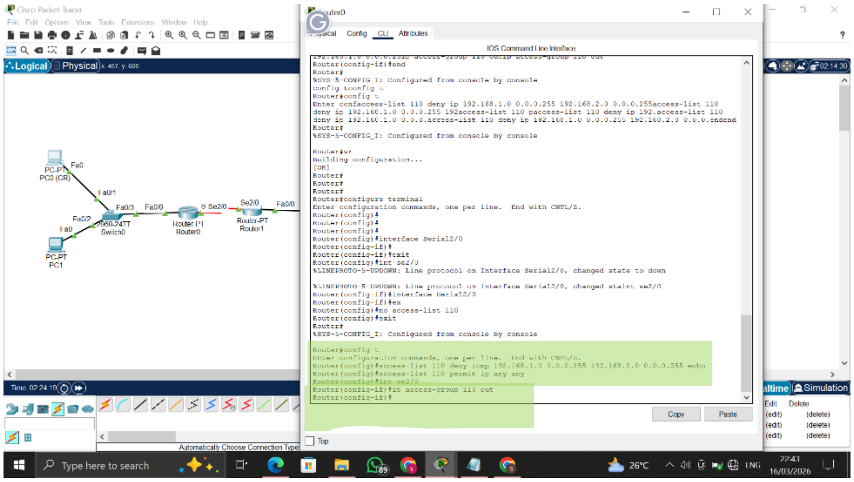

for more than 1 router -> routing is required
sucessfull packet-> msg send + responded
for static path req: dest network ip and its
router->config->static->req fill it and then a
next hop-> router that is directly connected
static routing -> routers should be direclty c
ACL (standard and extended):
ACL-> security for network based on some p
many types we'll cover standard and extende
policies required :
1-99: Standard
100-199: Extended
Router1
Physical Config CLI Attributes
Serial2/0
Port Status
Duplex
Clock Rate 1200
IP Configuration
IPv4 Address 192.168.3.2
Subnet Mask 255.255.255.0
Tx Ring Limit 10
Equivalent IOS Commands
00:06:31
Type here to search
25°C Partly cloudy
16:58
16/03/2026

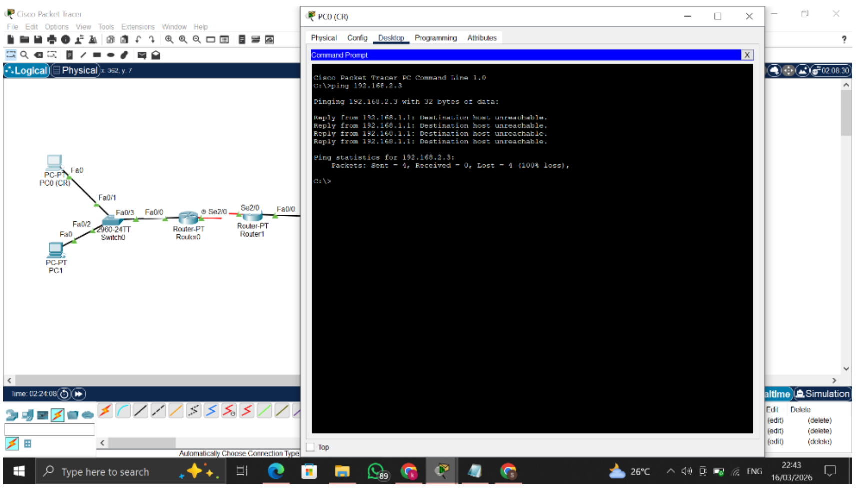

for more than 1 router -> routing is required
sucessfull packet-> msg send + responded
for static path req: dest network ip and its
router->config->static->req fill it and then a
next hop-> router that is directly connected
static routing -> routers should be direclty c
ACL (standard and extended):
ACL-> security for network based on some p
many types we'll cover standard and extende
policies required :
1-99: Standard
100-199: Extended
Router1
Physical Config CLI Attributes
FastEthernet0/0
Port Status
Bandwidth
Duplex
MAC Address 0000.0C24.E0D0
IP Configuration
IPv4 Address 192.168.2.1
Subnet Mask 255.255.255.0
Tx Ring Limit 10
Equivalent IOS Commands
00:06:28
Type here to search
25°C Partly cloudy
16:58
16/03/2026

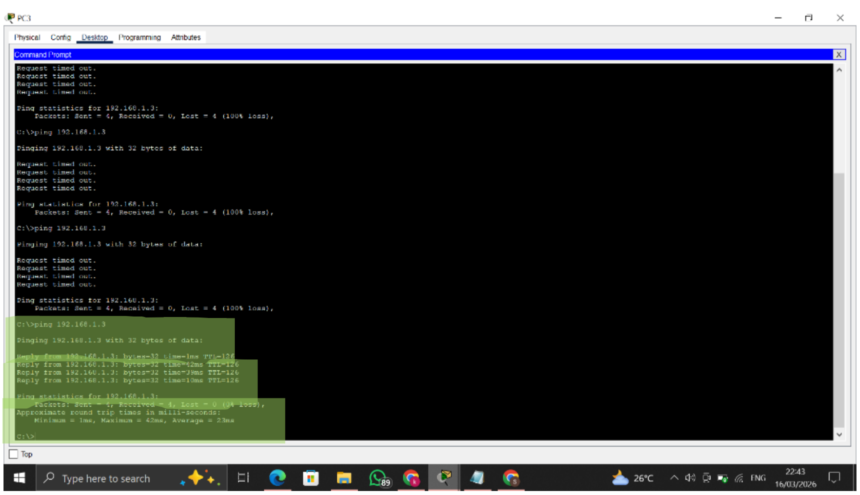

PC1
Physical
Config
Desktop
Programming
Attributes
GLOBAL
Settings
Algorithm Settings
INTERFACE
FastEthernet0
Bluetooth
FastEthernet0
Port Status
Bandwidth
Duplex
MAC Address
00D0.D314.AD8E
IP Configuration
DHCP
Static
IPv4 Address
192.168.1.3
Subnet Mask
255.255.255.0
IPv6 Configuration
Automatic
Static
IPv6 Address
Link Local Address
exp(InLab) - Notepad
for more than 1 rou
sucessfull packet->
for static path req:
router->config->sta
next hop-> router t
static routing -> ro
ACL (standard and
ACL-> security for
many types we'll co
policies required :
1-99: Standard
100-199: Extended
00:06:16
Realtime
Simulation
Type here to search
25°C Partly cloudy
ENG
16:58
16/03/2026


Router1
Physical
Config
CLI
Attributes
GLOBAL
Settings
Algorithm Settings
ROUTING
Static
RIP
INTERFACE
FastEthernet0/0
FastEthernet1/0
Serial2/0
Serial3/0
FastEthernet4/0
FastEthernet5/0
Static Routes
Network
Mask
Next Hop
Add
Network Address
192.168.1.0/24 via 192.168.3.1
Remove
Equivalent IOS Commands
for more than 1 router -> routi
sucessfull packet-> msg send +
for static path req: dest networ
router->config->static->req fill
next hop-> router that is direct
static routing -> routers should
ACL (standard and extended):
ACL-> security for network bas
many types we'll cover standar
policies required :
1-99: Standard
100-199: Extended
00:07:21
Realtime
Simulation
Type here to search
25°C Partly cloudy
ENG
16:59
16/03/2026

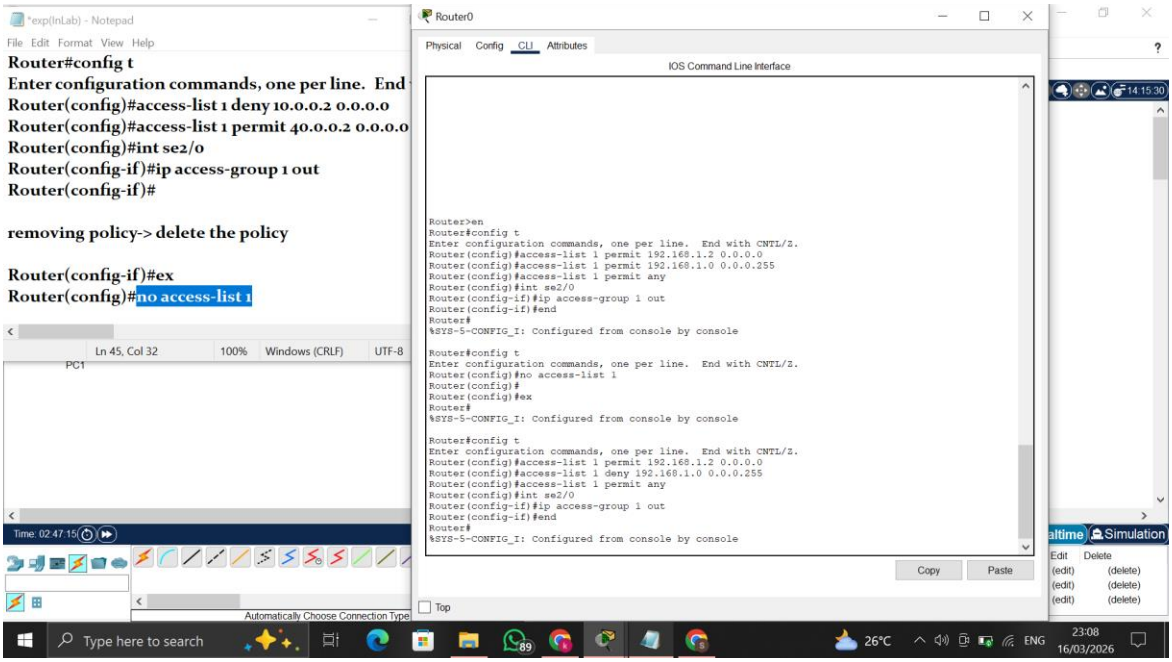

Router0
Physical
Config
CLI
Attributes
GLOBAL
Settings
Algorithm Settings
ROUTING
Static
RIP
INTERFACE
FastEthernet0/0
FastEthernet1/0
Serial2/0
Serial3/0
FastEthernet4/0
FastEthernet5/0
Static Routes
Network
Mask
Next Hop
Add
Network Address
192.168.2.0/24 via 192.168.3.2
Remove
Equivalent IOS Commands
for more than 1 router -> rout
sucessfull packet-> msg send
for static path req: dest netwo
router->config->static->req fi
next hop-> router that is direc
static routing -> routers shou
ACL (standard and extended):
ACL-> security for network ba
many types we'll cover standa
policies required :
1-99: Standard
100-199: Extended
00:07:11
Realtime
Simulation
Type here to search
25°C Partly cloudy
ENG
16:59
16/03/2026

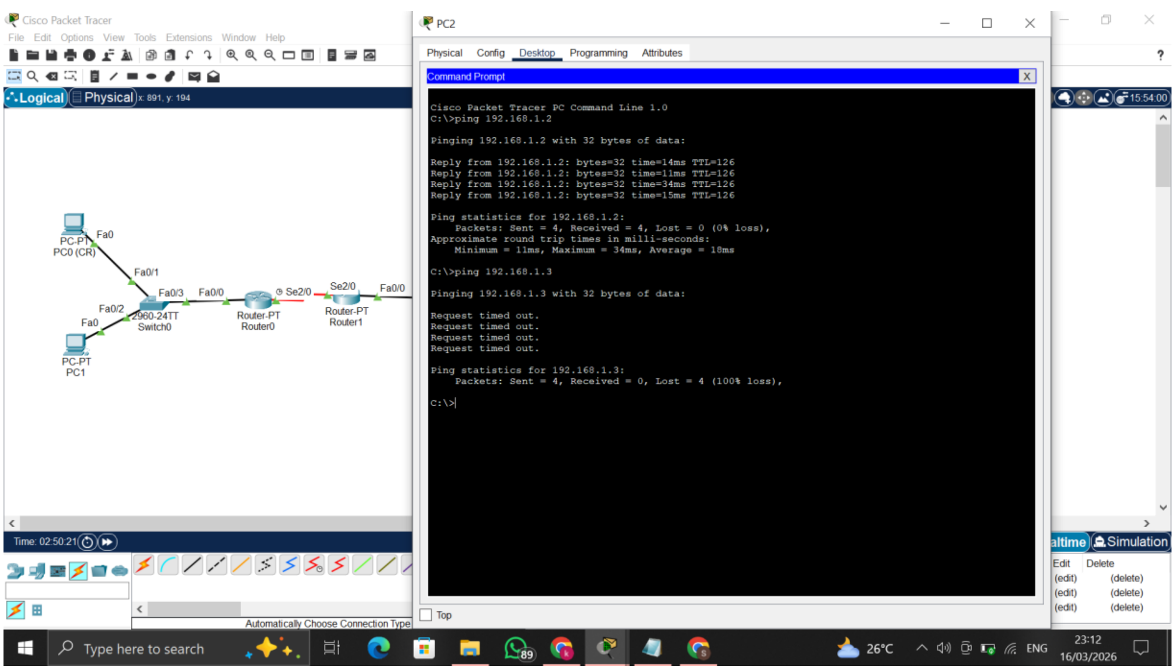

Router0
Config
CLI
Attributes
IOS Command Line Interface
Router(config-if)#end
Router#
%SYS-5-CONFIG_I: Configured from console by console
Router#config t
Router#
%SYS-5-CONFIG_I: Configured from console by console
Router#wr
Building configuration...
[OK]
Router#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#interface Serial2/0
Router(config-if)#exit
Router(config)#int se2/0
%LINEPROTO-5-UPDOWN: Line protocol on Interface Serial2/0, changed state to down
Router(config-if)#interface Serial2/0
Router(config-if)#ex
Router(config)#no access-list 110
Router(config)#exit
Router#
%SYS-5-CONFIG_I: Configured from console by console
Router#config t
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#access-list 110 deny icmp 192.168.1.0 0.0.0.255 192.168.2.0 0.0.0.255 echo
Router(config)#access-list 110 permit ip any any
Router(config)#int se2/0
Router(config-if)#ip access-group 110 out
Router(config-if)#
Copy
Paste
Top

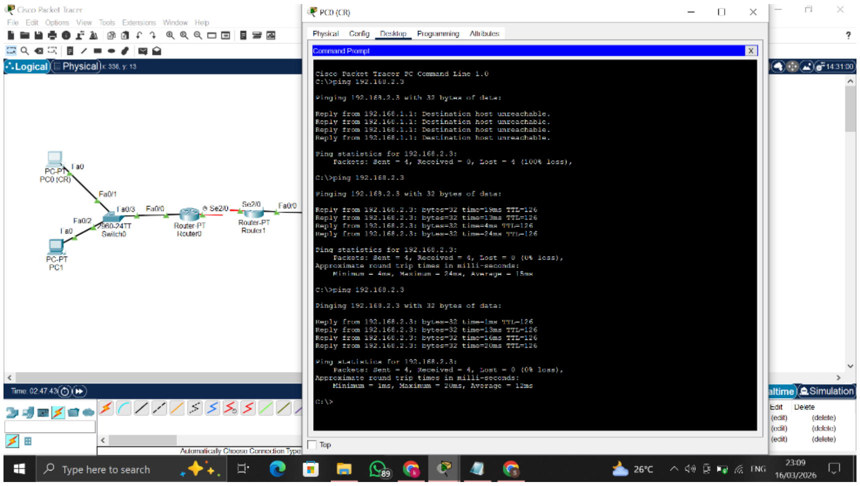

PC0 (CR)
Physical
Config
Desktop
Programming
Attributes
Command Prompt
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 192.168.2.3
Pinging 192.168.2.3 with 32 bytes of data:
Reply from 192.168.1.1: Destination host unreachable.
Reply from 192.168.1.1: Destination host unreachable.
Reply from 192.168.1.1: Destination host unreachable.
Reply from 192.168.1.1: Destination host unreachable.
Ping statistics for 192.168.2.3:
Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),
C:\>
Top

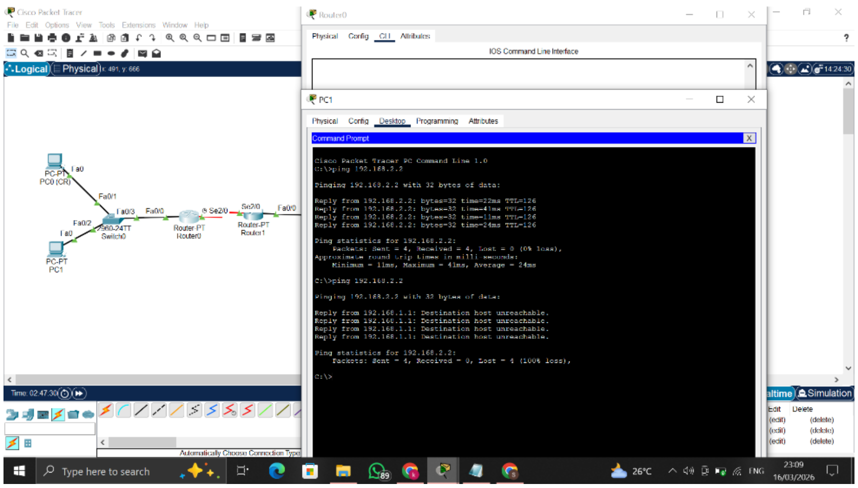

PC3
Physical
Config
Desktop
Programming
Attributes
Command Prompt
Request timed out.
Request timed out.
Request timed out.
Request timed out.
Ping statistics for 192.168.1.3:
Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),
C:\>ping 192.168.1.3
Pinging 192.168.1.3 with 32 bytes of data:
Request timed out.
Request timed out.
Request timed out.
Request timed out.
Ping statistics for 192.168.1.3:
Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),
C:\>ping 192.168.1.3
Pinging 192.168.1.3 with 32 bytes of data:
Request timed out.
Request timed out.
Request timed out.
Request timed out.
Ping statistics for 192.168.1.3:
Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),
C:\>ping 192.168.1.3
Pinging 192.168.1.3 with 32 bytes of data:
Reply from 192.168.1.3: bytes=32 time=42ms TTL=126
Reply from 192.168.1.3: bytes=32 time=10ms TTL=126
Approximate round trip times in milli-seconds:
Minimum = 1ms, Maximum = 42ms, Average = 23ms
C:\>
Top
Type here to search

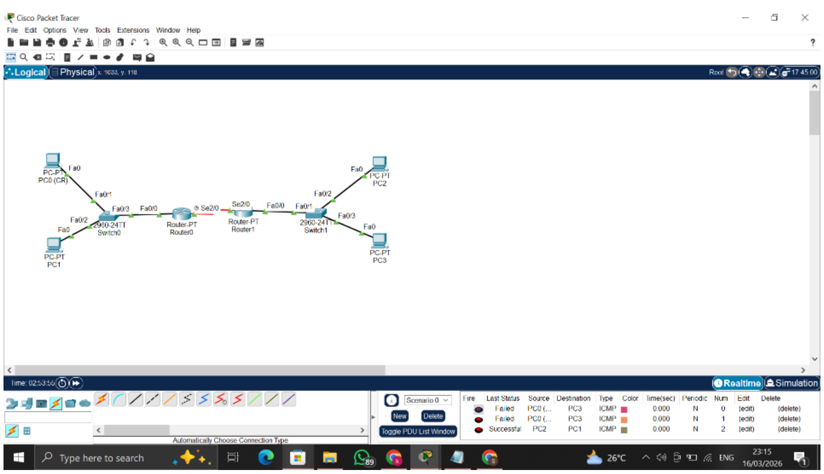
2. *Only allow any one PC in the STUDENTS network (for example: as a CR communicates with teachers) to access and communicate with the TEACHERS network, blocking every other device in the STUDENTS network.*






PC0 (CR)
Physical Config Desktop Programming Attributes
Command Prompt
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 192.168.2.3
Pinging 192.168.2.3 with 32 bytes of data:
Reply from 192.168.1.1: Destination host unreachable.
Reply from 192.168.1.1: Destination host unreachable.
Reply from 192.168.1.1: Destination host unreachable.
Reply from 192.168.1.1: Destination host unreachable.
Ping statistics for 192.168.2.3:
Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),
C:\>ping 192.168.2.3
Pinging 192.168.2.3 with 32 bytes of data:
Reply from 192.168.2.3: bytes=32 time=19ms TTL=126
Reply from 192.168.2.3: bytes=32 time=13ms TTL=126
Reply from 192.168.2.3: bytes=32 time=4ms TTL=126
Reply from 192.168.2.3: bytes=32 time=24ms TTL=126
Ping statistics for 192.168.2.3:
Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
Minimum = 4ms, Maximum = 24ms, Average = 15ms
C:\>ping 192.168.2.3
Pinging 192.168.2.3 with 32 bytes of data:
Reply from 192.168.2.3: bytes=32 time=1ms TTL=126
Reply from 192.168.2.3: bytes=32 time=13ms TTL=126
Reply from 192.168.2.3: bytes=32 time=16ms TTL=126
Reply from 192.168.2.3: bytes=32 time=20ms TTL=126
Ping statistics for 192.168.2.3:
Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
Minimum = 1ms, Maximum = 20ms, Average = 12ms
C:\>


IOS Command Line Interface
PC1
Physical Config Desktop Programming Attributes
Command Prompt
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 192.168.2.2
Pinging 192.168.2.2 with 32 bytes of data:
Reply from 192.168.2.2: bytes=32 time=22ms TTL=126
Reply from 192.168.2.2: bytes=32 time=41ms TTL=126
Reply from 192.168.2.2: bytes=32 time=11ms TTL=126
Reply from 192.168.2.2: bytes=32 time=24ms TTL=126
Ping statistics for 192.168.2.2:
Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
Minimum = 11ms, Maximum = 41ms, Average = 24ms
C:\>ping 192.168.2.2
Pinging 192.168.2.2 with 32 bytes of data:
Reply from 192.168.1.1: Destination host unreachable.
Reply from 192.168.1.1: Destination host unreachable.
Reply from 192.168.1.1: Destination host unreachable.
Reply from 192.168.1.1: Destination host unreachable.
Ping statistics for 192.168.2.2:
Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),
C:\>


Cisco Packet Tracer